Dimensional transmutation in a hidden SU(3) sector: the ghost dark-energy scale μ ≈ 28.5 MeV and glueball dark matter

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Abstract

The QCD-ghost (Veneziano contact-term) dark-energy law $\rho = \mu^3 H$ requires a mass scale $\mu \approx 28.5$ MeV. We ask whether μ is the confinement scale of a hidden $SU(3)$ sector fixed by dimensional transmutation. A pure hidden Yang–Mills sector confining at μ requires a Planck-scale coupling $\alpha_h = 1/83.2$: log-natural, replacing a tuning of one part in 10^{60}, though the exponential map retains a sensitivity of 47.5. A sharper variant, *mirror unification* — hidden and visible couplings coinciding at the Planck scale, with six flavours — introduces zero new continuous parameters and lands in the right decade: the loop and scheme band is 28–105 MeV at two loops, widening to 28–133 MeV through four-loop running. The same sector's glueballs are viable dark matter through $3 \to 2$ cannibal freeze-out — a mass miracle, not an abundance one — with a velocity-independent self-interaction. Cluster bounds and a conditional Little-Red-Dot seeding argument converge, as constraints, near 52 MeV. That scale and the 28.5 MeV amplitude scale are one sector, related by the contact-term coefficient $6^{1/3}$: their cubed ratio of 6 is a predicted, lattice-testable topological factor rather than an independent tuning, to the accuracy of the pure-glue relations it rests on. Three limits are explicit. The dark-energy law is inherited from a contact term that remains contested and unproven, and is the single load-bearing conjecture. Near the conformal-window edge the perturbative band is irreducibly order-unity wide, so separating the coefficient choices is a lattice question. Every claim is tagged derived, matched, or conjectured.

Keywords: dark matter, hidden sector, dimensional transmutation, glueball

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dark matter, self-interacting dark matter, dark energy

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1. Introduction

The QCD-ghost, or Veneziano contact-term, construction gives a dark-energy density linear in the Hubble rate, $\rho_{\mathrm{DE}} \propto \Lambda^3 H$ [1, 2, 3, 4]. Taken with the visible-sector scale Λ_{QCD} it overshoots the observed dark-energy density, and the construction is usually left there. Substituting a *hidden* confinement scale instead,

$$\rho_X = \mu^3 H, \qquad \mu \approx 28.5\,\text{MeV}, \tag{1}$$

reproduces the observed amplitude — but only by trading one unexplained number for another, since 28.5 MeV put in by hand is a vacuum energy tuned to one part in $\sim 10^{60}$ against the Planck scale. The question this paper asks is whether that scale can instead be *generated*.

Dimensional transmutation [5] is the natural candidate, and the only known mechanism converting an $O(1)$ dimensionless input into an exponentially suppressed dimensionful output without tuning:

$$\Lambda = M_{\mathrm{Pl}}\, e^{-2\pi/(b_0\,\alpha(M_{\mathrm{Pl}}))}. \tag{2}$$

We assess the proposal that μ is the confinement scale Λ_h of a hidden $SU(3)$ gauge sector, in two versions of increasing economy — a postulated coupling, then a *mirror-unified* coupling with no new continuous input — together with the mechanism that would convert the *scale* into the *law* (the Urban–Zhitnitsky/Ohta topological vacuum energy), and the observation that the same sector's glueballs are a viable dark-matter candidate with distinctive, velocity-independent self-interactions.

The law itself is not new: it is the established contact-term result with the single substitution $\Lambda_{\mathrm{QCD}} \to \Lambda_{\mathrm{hidden}}$. Nor is glueball dark matter new [6, 7, 8, 9]. The contribution here is (i) fixing μ by dimensional transmutation of a hidden $SU(3)$, and (ii) tying that same sector to glueball dark matter, so that one confinement scale carries both roles. Neither ingredient is new on its own; what is new is that they share a scale. Notably the same H-linear form is reached from an entirely different route — a volume-law horizon entropy gives $\rho \propto H$ in entropic cosmology [10, 11] — so $\rho \propto H$ is doubly motivated, by the QCD contact term *and* by horizon thermodynamics, a convergence rather than a coincidence. The QCD-topological-sector program remains active and is being tested against current cosmological data [12].

The contact term enters only in §4, and it is worth separating at the outset what does not depend on it. The transmutation calculation, the mirror-unification band and the glueball relic analysis take as input only the value of μ, standard gauge-theory running, and lattice pure-glue relations; none of them uses the contact term. Should the Urban–Zhitnitsky mechanism fail, what remains is a hidden $SU(3)$ sector confining at 30–130 MeV with no new continuous parameters, whose glueballs are a viable dark-matter candidate with a velocity-independent self-interaction; a negative result — that no perturbative order can discriminate the two candidate coefficients (§3.3); and a cosmology-free lattice test that decides the identification either way (§4.3). What would be lost is the *dark-energy interpretation* of that sector, not the sector itself: the contact term is the one conjectural load here, and it does not carry the rest.

The H-linear dark-energy phenomenology that motivates the scale — the equation of state, the growth gate and the modified-growth signatures — is developed in a companion cosmology manuscript, which identifies the origin of μ as its deepest open problem. Nothing below depends on that development.

Each claim below is tagged by provenance: which quantities are *derived* (following from stated inputs by a reproducible calculation), which are *matched* to data (a computed number compared against an observed or required value), which are $O(1)$-controlled *estimates*, and which are *conjectures* (a mechanism whose validity is not established). Every numerical claim below carries its provenance; all are reproducible from the scripts listed in §10.

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2. The transmutation mechanism

2.1. The coupling that does the job

For a pure hidden $SU(3)$ Yang–Mills sector ($b_0 = 11$), one-loop transmutation from the Planck scale gives

$$\frac{1}{\alpha_h(M_{\rm Pl})} = \frac{11}{2\pi} \ln \frac{M_{\rm Pl}}{\Lambda_h}. \tag{3}$$

Setting $\Lambda_h = \mu = 28.55$ MeV yields

$$\frac{1}{\alpha_h(M_{\rm Pl})} = 83.2, \qquad \alpha_h(M_{\rm Pl}) = 0.0120. \tag{4}$$

This is the same ballpark as the Standard Model couplings at $M_{\rm Pl}$ ($1/\alpha \sim 47$–59), though not equal to any of them. The check verifies exactly (`src/mu_transmutation_check.py`). The 10^{60} tuning in ρ genuinely becomes a log-natural coupling specification.

We quote here the full Planck mass $M_{\rm Pl} = 1.22 \times 10^{19}$ GeV; the reduced-mass convention gives $1/\alpha \approx 80$. We therefore quote $1/\alpha_h(M_{\rm Pl}) \approx 80$–$83$, a convention band rather than a determination.

2.2. Sensitivity of the transmutation map

The transmutation map is exponential; its Jacobian is

$$\frac{d \ln \Lambda_h}{d \ln \alpha_h} = \ln \frac{M_{\rm Pl}}{\Lambda_h} = 47.5. \tag{5}$$

A 1% specification of $\alpha_h(M_{\rm Pl})$ fixes μ only to $\sim 48\%$ (a factor 1.6); pinning μ to 1% would require α_h specified to 0.02%. The correct statement is therefore: transmutation naturalizes the scale; the precise value of μ remains a mild, log-level selection unless the coupling itself is fixed by a principle. That principle is the subject of §3.

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3. Mirror unification: zero new continuous parameters in, an O(1) band out

3.1. The construction

Rather than postulating α_h, demand that the hidden coupling *equal the visible QCD coupling at the Planck scale* — the coupling-inheritance structure familiar from mirror/twin constructions [13]:

$$\alpha_h(M_{\rm Pl}) = \alpha_s(M_{\rm Pl}). \tag{6}$$

Running the measured $\alpha_s(M_Z) = 0.1181$ up at two loops (with the $N_f\ 5 \to 6$ threshold at m_t) gives $\alpha_s(M_{\rm Pl}) = 0.01886$, i.e. $1/\alpha = 53.0$. The hidden confinement scale is then fixed entirely by the *discrete* flavor content of the mirror sector. At one loop (`src/mu_transmutation_check.py`):

hidden $SU(3)$ content	Λ_h (one loop)
$N_f = 0$	1.2×10^6 MeV
$N_f = 3$	1.6×10^3 MeV
$N_f = 6$	$\sim$30–46 MeV
$N_f = 7$	0.3 MeV

A mirror QCD with six light flavors and a Planck-unified coupling confines within a factor ~ 2 of μ, with no new dimensionless inputs: the coupling is inherited and N_f is discrete. This is the strongest form of the proposal, and the "zero new continuous parameters" claim attaches precisely here — to the input. It does *not* attach to the output, as the next two subsections make quantitative.

3.2. Two-loop refinement: the single number dissolves into a band

The two-loop calculation with thresholds (`src/mu_transmutation_steps.py`) supersedes the early one-loop quote of "$\Lambda_h \approx 46$ MeV":

scheme / order	Λ_h
1-loop pole	27.6 MeV
2-loop pole	105 MeV
2-loop $\overline{\text{MS}}$-form	88 MeV
(targets)	bare $\mu = 28.5$ · UZ-corrected $6^{1/3}\mu = 51.9$

Two conclusions follow. (i) Mirror unification with $N_f = 6$ robustly lands in the right decade with zero new continuous parameters — a consistency check that the mirror hypothesis is not excluded. $N_f = 6$ is the flavor content selected to land in the decade ($N_f = 5, 7$ miss by orders of magnitude); given $\alpha_s(M_{\text{Pl}}) \sim 1/53$ and $d \ln \Lambda / d \ln \alpha \approx 47.5$, hitting the decade was near-automatic — a consistency check, not a prediction. (ii) The loop/scheme spread ($\times 2$ each way, band $\approx$ 28–105 MeV) cannot distinguish the bare coefficient ($\Lambda_h = \mu$) from the UZ-corrected one ($\Lambda_h = 6^{1/3}\mu$, §4). The band brackets *both* targets, so no alignment claim can be drawn from it.

3.3. The three-loop result: the band does not tighten

The scheme-fixed three-loop (and four-loop) running was computed as Tier-1 item #2 (`src/mirror_qcd_3loop.py`), with the explicit question of whether it tightens the band. It does not:

loop order	Λ_{pole}
1	28 MeV
2	105 MeV
3	spurious IR fixed point — β develops a zero at $a \approx 0.28$ ($\alpha \approx 3.5$), outside perturbative control
4	133 MeV

The three-loop does not tighten the band; it reveals why no perturbative order will. $N_f = 6$ $SU(3)$ sits close enough to the conformal-window edge ($N_f^* \sim 8$–12) that the confinement scale is loop-order-sensitive: the 1-loop–4-loop span is a factor ~ 5 (28–133 MeV), and the three-loop β-function's fixed point at $\alpha \approx 3.5$ is an artifact appearing where perturbation theory has already failed. The band is irreducibly $O(1)$-wide from perturbation theory near the conformal edge. This is a negative refinement of the two-loop result and we present it as such: the zero-parameter mirror construction brackets μ to the right decade — a real success — but perturbation theory *cannot* pin the value to 28.5 vs 52 MeV (Figure 1).

3.4. The closer: lattice $T_c/\Lambda_{\overline{\text{MS}}}$ [SPECIFIED, external]

The physical scale requires one nonperturbative $O(1)$ input: the ratio $T_c/\Lambda_{\overline{\text{MS}}}$ for $N_f = 6$ $SU(3)$. Per external-calculation spec S-4, this is a pure lattice question with standard technology (light degenerate quarks, T_c from the chiral-susceptibility peak, $\Lambda_{\overline{\text{MS}}}$ via the gradient-flow coupling); existing near-conformal-window $N_f \approx 6$ studies may permit a reanalysis without new runs. The pre-registered decision rule: $T_c/\Lambda_{\overline{\text{MS}}}$ pins Λ_h within $\sim 15\%$, at which point the transmutation μ is either 28.5 MeV (bare) or 52 MeV (UZ-corrected) — discriminating the two coefficients that the astrophysical probes of §7 attack from opposite sides.

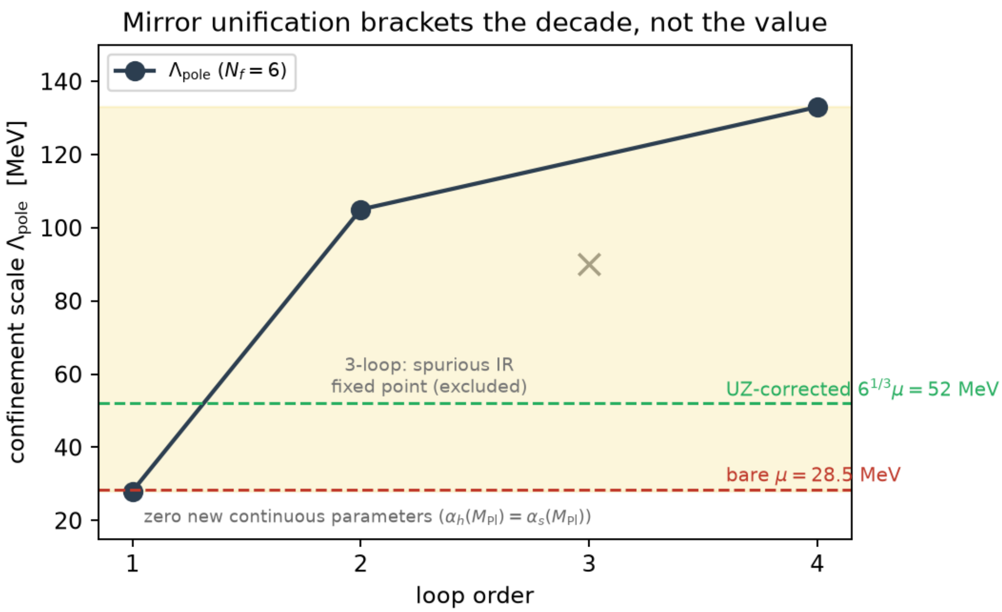

Figure 1: Mirror unification brackets the decade, not the value. The confinement scale $\Lambda_{\rm pole}$ from $N_f = 6$ running, with zero new continuous parameters ($\alpha_h(M_{\rm Pl}) = \alpha_s(M_{\rm Pl})$), spans 28–133 MeV across loop orders (the three-loop point is a spurious IR fixed point where perturbation theory has already failed). The band is irreducibly $O(1)$-wide near the conformal-window edge and cannot discriminate the bare $\mu = 28.5$ MeV from the UZ-corrected $6^{1/3}\mu = 52$ MeV — a lattice question (§3.4).

4. From scale to law: the Urban–Zhitnitsky mechanism (conjectured)

The law $\rho_X = \mu^3 H$ is not itself new: it is the Urban–Zhitnitsky/Ohta QCD-ghost (Veneziano contact-term) dark-energy law $\rho_{\mathrm{DE}} \propto \Lambda^3 H$ [1, 2, 3, 4, 14], with the single substitution $\Lambda_{\mathrm{QCD}} \to \Lambda_{\mathrm{hidden}}$. Our contribution is not the law but (i) fixing the scale μ by dimensional transmutation of a hidden $SU(3)$, and (ii) tying the same sector to glueball dark matter.

4.1. The mechanism and its contested status

Transmutation supplies the *scale*; SEDE needs the *law* $\rho_X = \mu^3 H$. The candidate bridge is the Urban–Zhitnitsky/Ohta proposal [1, 2, 15, 3]: in a universe with a finite causal horizon, the topological (Veneziano-ghost / contact-term) sector of a confining theory shifts the vacuum energy from its Minkowski value by

$$\Delta\rho \sim H \cdot \frac{\chi_{\mathrm{top}}}{m_G} \sim \frac{H\,\Lambda_h^3}{6}, \tag{7}$$

using the lattice pure-glue relations $m_G \approx 6\Lambda$ and $\chi_{\mathrm{top}} \sim \Lambda^4$. The linear-in-$H$, boundary-sensitive form is exactly SEDE's law, and it fixes the required confinement scale not at μ but at

$$\Lambda_h = 6^{1/3}\mu = 52\ \mathrm{MeV}. \tag{8}$$

Status: conjectural and contested. The Veneziano ghost is gauge-variant and non-propagating; whether the linear-in-H term survives careful regularization has been debated since ∼2011 and is unsettled in the literature [16, 17, 18]. Within the transmutation route this is the single load-bearing assumption: if the UZ term does not survive, the route delivers a natural scale with no mechanism for the law.

4.2. The structural argument: the contact term as a candidate identity for the nonlocal degrees of freedom

There is nonetheless a structural reason to take the mechanism seriously, and it does not depend on accepting the contact term. A volume-law horizon entropy *cannot* be the entanglement entropy of local quantum-field degrees of freedom: the Casini/Bekenstein bound gives $S_{\mathrm{vol}}/S_{\mathrm{Bek}} = 1/H_0R$, so a volume-law count necessarily forces nonlocal horizon degrees of freedom. Identifying them is the central open structural problem for any construction of this kind. The Veneziano-ghost/topological sector is precisely a nonlocal, non-propagating, boundary-sensitive contact term, not a local propagating QFT mode [19, 20, 21]. The UZ mechanism is therefore one candidate in a class of horizon-sensitive vacuum-energy mechanisms [1, 2, 3, 4, 22, 23], all sharing the same regularization controversy; its specific appeal here is the χ_{top}/m_G coefficient, not uniqueness. It simultaneously (a) produces $\rho \propto H\Lambda^3$ and (b) has the nonlocal character the entropy count demands. It also *evades* (does not violate) the Casini bound applied at all times, which excludes local processes from producing the horizon count: the contact-term vacuum energy is not local entanglement by construction. The transmutation proposal is thus not only a candidate origin for μ; it is a candidate identity for those nonlocal degrees of freedom. This is an alignment of structures, not a derivation, and the mechanism remains a conjecture.

The gate also fits structurally: UZ supplies the *ceiling* $\mu^3 H$, and SEDE's activation fraction $u = f_{\text{sat}} \in [0, 1]$ (the foundations paper's "occupancy" reading) supplies the timing, driven by the Avrami structure-collapse coverage. What makes the contact term *gateable* remains an open question — a covariant formulation of the gate — unchanged by this proposal.

4.3. The lattice discharge test [SPECIFIED, external]

The contested-mechanism question can be converted into a computable yes/no with no cosmology at all. The UZ claim implies a finite-volume/boundary dependence of the pure-glue vacuum energy:

$$\rho_{\text{vac}}(L) - \rho_{\text{vac}}(\infty) \propto \frac{\chi_{\text{top}}}{m_G\, L}, \tag{9}$$

which is lattice-measurable in principle (pure $SU(3)$, varying box size and boundary conditions, target coefficient χ_{top}/m_G). This is the sharpest available discharge path for the load-bearing conjecture.

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5. Glueball dark matter

5.1. Cannibal freeze-out and the natural coupling

The hidden sector confines into glueballs of mass $m_G = 6\Lambda_h \approx 180$–$630$ MeV across the perturbative band. Glueballs of a hidden non-Abelian sector are a well-studied dark-matter candidate [6, 7, 8, 24], with the relic abundance and ΔN_{eff} set by the confining sector's cosmology [9, 25]. Neither the candidate nor the cannibal relic mechanism is new here. What is specific to this construction is that the confinement scale is not a free parameter: it is the same Λ_h fixed in §3 by transmutation and, through the contact-term coefficient, tied to the dark-energy amplitude. The dark-matter mass and the dark-energy scale are then not independent inputs but one number in two roles, $m_G = 6\Lambda_h = 6^{4/3}\mu$ — a relation that is lattice-testable (§4.3) and that fails if either side moves. Pure glue has no renormalizable portal to the Standard Model, so the operative relic regime is the decoupled cannibal one [26]: number-changing $3 \to 2$ (SIMP-type) self-annihilations [27, 28, 29] with separate hidden-sector entropy, frozen out when $\Gamma_{3\to 2} = H$ [30, 31, 32]. The full calculation (`src/glueball_simp_relic.py`: separate-entropy tracking, Boltzmann-equilibrium cannibalization) confirms the coupling naturalness: solving for $\Omega h^2 = 0.12$ requires

$$\alpha_{\text{eff}} \approx 0.25\text{–}0.6 \tag{10}$$

at the viable points — exactly what a confining sector at its own scale supplies. This is a value matched to the required relic abundance, not an independent prediction.

An earlier exclusion is retracted for this regime: the "$\Delta N_{\text{eff}} = 0.43 \Rightarrow$ marginally excluded" result applied to a hidden sector still *relativistic* at BBN/CMB. Here confinement at $T_h \sim \Lambda_h$ converts the gluons to massive glueballs long before BBN and cannibalization keeps them non-relativistic: $\Delta N_{\text{eff}} \approx 0$, invisible to CMB-S4. The dark-radiation channel is therefore *not* the test of this sector — a testability inversion that the signatures of §7 replace.

For $\xi \approx 0.005$ the hidden sector confines and becomes non-relativistic well before BBN, so it is cold rather than warm; the cannibal $3 \to 2$ heating [31, 30] does not spoil this at such small ξ.

5.2. A mass miracle, not an abundance miracle

The framing "SIMP freeze-out lands on Ω_{DM} with no fine-tuning" holds for the *mass scale and coupling* but not for the abundance. In the decoupled regime the relic is controlled by the hidden-to-visible temperature ratio $\xi = T_h/T_{\text{vis}}$, and the calculation gives a sharp sensitivity statement:

> Sweeping α_{eff} across the entire natural range $[0.5, 4\pi]$ moves the viable ξ by only $\times 1.11$. The abundance requires $\xi = 0.0062$–0.0069 ($\Lambda_h = 30$ MeV) / 0.0051–0.0057 (52 MeV) / 0.0040–0.0044 (105

MeV) — an ~11%-wide band. Since $Br \propto \xi^4$, the inflaton branching ratio to the hidden sector must be $Br \approx (0.4\text{–}3) \times 10^{-10}$, selected to $\sim 40\%$ — 3.5 decades above pure gravitational leakage ($Br_{\text{grav}} \approx 7 \times 10^{-14}$) and 10 decades below thermal contact.

This proposal trades ECCG-ADM's *explained* $\Omega_{\text{DM}}/\Omega_b \approx 5$ for a $Br \sim 2 \times 10^{-10}$ reheating selection — one dial for another. The new dial is arguably more natural (a tiny branching to a portal-less sector is generic; the $\times 1.11$ band within it is not), but the "why comparable to baryons" coincidence returns unexplained. The SIMP miracle here is a mass miracle — m_G lands where $3 \to 2$ works with a natural coupling, tied to the dark-energy scale — not an abundance miracle.

5.3. *A theorem-level observation*

A single-source constraint ("η_B-calculable XOR m_X-sharp") holds for *clock-sourced* dark matter — a construction can deliver a calculable η_B or a sharp m_X, but not both — with the premise that both yields ride the clock's $\mu = H/2\pi$. Glueball dark matter is *thermally* sourced, so the premise fails, and this branch achieves both a derived η_B (through the Candidate-C clock channel) and a predicted DM mass — the first branch of the framework to do so. The theorem's content is conserved, relocated from the mass to the abundance: exactly the Br dial of §5.2.

5.4. *The subcomponent corner and the m_X-shift signature*

If instead the ADM relic remains the dominant dark matter (the original ECCG branch) and the glueball is a subcomponent, the boundary structure at $\Lambda_h = 52$ MeV ($m_G \approx 312$ MeV, $\Omega_G/\Omega_{\text{DM}} \propto \xi^3$) is:

regime	$\xi = T_h/T_{\text{vis}}$
invisible ($\Omega_G < 0.1\%$ of DM)	$\xi < 0.0006$
absorbable subcomponent (m_X shifts *within* ECCG's quoted 1.63–1.78 GeV band)	$0.0006 < \xi < 0.0025$
branch transition (ADM-dominant $\to$ glueball-dominant world)	$\xi > 0.0025$
overclosed	$\xi > 0.0058$

The absorbable window carries a real signature: a future m_X measurement at the *low* end of the band (1.63–1.70 GeV) would hint at a glueball subcomponent — the transmutation sector becoming indirectly visible through the dark-matter mass rather than through ΔN_{eff}.

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6. Signatures: velocity-independent SIDM and the pincer on $\boldsymbol{\Lambda_h}$

6.1. Self-interaction fixed by the scale

Glueball dark matter self-interacts with a geometric, velocity-independent cross-section fixed by Λ_h alone (`src/glueball_simp_relic.py`):

Λ_h	σ/m
30 MeV	0.47 cm^2/g
52 MeV (UZ)	0.09 cm^2/g
105 MeV	0.01 cm^2/g

(These full-calculation values supersede an interim estimate of $\approx\ 0.26$ cm^2/g at 52 MeV quoted in the steps note.) Velocity-independence is a simplifying strength — there is no $\sigma(v)$ freedom to adjust — which also means the *same* σ/m must satisfy dwarf and cluster constraints simultaneously. At $m_G\ \sim\ 300$ MeV the de Broglie wavelength is microscopic, so no superfluid/condensate halo phenomenology arises.

Cluster bounds press from above. Cluster-scale SIDM constraints at $\sigma/m \lesssim 0.1$–0.5 cm^2/g [33, 34, 35] probe the low-Λ_h end directly: an exclusion of $\sigma/m\ >\ 0.1$ pushes the sector to $\Lambda_h\ \gtrsim\ 50$ MeV — i.e. it *selects the UZ-corrected coefficient over the bare one*. Halo astrophysics would, in that event, be measuring the topological-susceptibility coefficient of §4.

6.2. Gravothermal seeding of the Little Red Dots presses from below (derived, conditional)

SIDM gravothermal collapse of early dense halos is a heavy-seed channel for the JWST Little Red Dot (LRD) overmassive black holes ($M_{\text{BH}}\ \sim\ 10^6$–$10^8\,M_\odot$ at $z\ \sim\ 5$–9, $n_{\text{LRD}}\ \sim\ 10^{-5}\,\text{Mpc}^{-3}$). The calculation (`src/glueball_` `lrd_seeding.py`: $\sigma(M,z)$ from the linear $P(k)$, Sheth–Tormen mass function, NFW gravothermal time with the Essig et al. calibration, lognormal concentration tail, formation window $z_f \in \{10\text{–}15\} \to z = 7$) gives, at the best formation redshift $z_f = 15$:

Λ_h	σ/m	DM-only $n_{\rm seed}/n_{\rm LRD}$	baryon-boosted $\times 30$ $n_{\rm seed}/n_{\rm LRD}$	seed mass
30 MeV	0.47	8×10^{-4} (needs $c > 30$)	5×10^2 (overshoot; absorbable by LRD duty cycle $\sim 10^{-2}$–10^{-3})	$\sim 5 \times 10^6$–$10^7\,M_\odot$
52 MeV (UZ)	0.09	2×10^{-8}	2.5 — crosses the LRD abundance	$\sim 7 \times 10^6\,M_\odot$ — in the LRD range
105 MeV	0.01	0 (no halo at any $c \leq 40$)	7×10^{-5}	—

Two findings:

1. DM-only, the glueball cannot seed the LRDs — robustly. Even the most-interacting case falls three orders of magnitude short at maximum optimism on every other axis; median-concentration collapse times are 90/470/4200 Gyr at $\Lambda_h = 30/52/105$ MeV.
2. With baryon-accelerated collapse (the $\times 30$ Feng/Yu-type bracket), $\Lambda_h = 52$ MeV lands within a factor ~ 2.5 of the LRD abundance with seed masses inside the observed range, $\Lambda_h = 30$ overshoots $\times 540$, and $\Lambda_h = 105$ fails by 10^4 even boosted.

Error budget. The seed abundance traverses ~ 6 orders of magnitude between the DM-only and $\times 30$-boosted brackets; the boost factor, the collapse-time calibration $C \approx 0.75$, the concentration parameters $\bar{c}$ and $\sigma_{\ln c}$, and the 2% seed fraction are each $O(1)$-uncertain, and the abundance responds *exponentially* through the concentration tail (which is trusted only to $\sim 4\sigma$, beyond which it is not simulation-calibrated). The "$\Lambda_h = 52$ lands on target" statement is therefore a viability statement, not a prediction: the claim is that *within the plausible baryon-boost bracket,* $\Lambda_h \approx 52$ *MeV crosses* n_{LRD} *while 105 cannot and 30 over-seeds.*

6.3. The pincer: one live bound and one conditional bracket (an estimate)

Putting §6.1 and §6.2 together with §4:

- clusters press from above: $\sigma/m \lesssim 0.1$–0.5 cm²/g disfavors $\Lambda_h \approx 30$ MeV;
- LRD seeding (if SIDM-driven) presses from below: $\sigma/m \gtrsim 0.1$ cm²/g (with baryons) excludes $\Lambda_h \approx 105$ MeV.

This is one live constraint (clusters, from above) plus one conditional viability bracket (LRD, from below), consistent with the UZ-defined 52 MeV — not three independent probes, since the UZ arm is the *definition* of 52 MeV ($= 6^{1/3}\mu$), not an independent measurement of it. The two-sided squeeze is consistent with $\Lambda_h \approx 52$ MeV — the same value the SEDE H-linear mechanism requires. This is a convergence of *constraints* (one of them conditional on an unresolved astrophysical question), not a measurement of Λ_h.** The lattice $T_c/\Lambda_{\overline{\mathrm{MS}}}$ calculation (§3.4) would supply a genuinely independent, non-astrophysical probe — either converging on one number, or failing to and falsifying the transmutation identification (Figure 2).

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7. Falsifiers

Numbered, with current status:

1. SIDM cluster discriminator (selects UZ vs bare). A cluster-scale exclusion of velocity-independent $\sigma/m > 0.1$ cm²/g pushes $\Lambda_h \gtrsim 50$ MeV, selecting the UZ-corrected coefficient over the bare one; conversely, a robust detection of velocity-*dependent* self-interaction, or of velocity-independent σ/m inconsistent with the whole 0.01–0.47 cm²/g ladder, cuts against the glueball horn. *Status: live; low-*Λ_h *end already at the constraint boundary.*
2. Low-end m_X window (subcomponent corner). In the ADM-dominant branch, a measured $m_X \in$ 1.63–1.70 GeV is the glueball-subcomponent signature; m_X pinned at the top of the ECCG band with no shift bounds $\xi < 0.0006$ and renders the subcomponent invisible. *Status: awaiting an* m_X *measurement.*

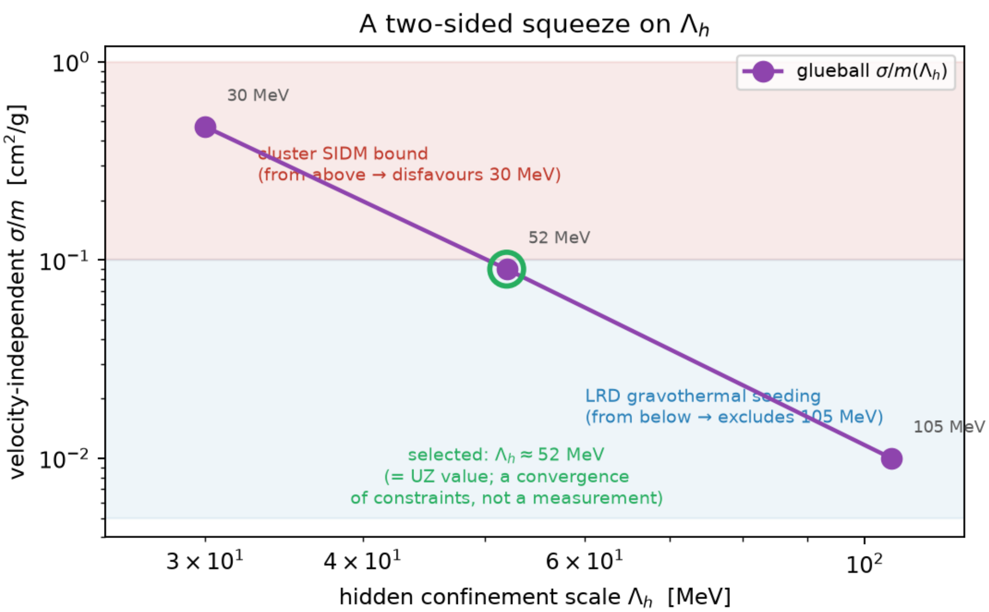

Figure 2: The two-sided squeeze on Λ_h. The velocity-independent glueball self-interaction σ/m falls monotonically with Λ_h (0.47 / 0.09 / 0.01 cm^2/g at 30 / 52 / 105 MeV). Cluster SIDM bounds press from above (disfavouring 30 MeV); Little-Red-Dot gravothermal seeding, if SIDM-driven, presses from below (excluding 105 MeV). The window selects $\Lambda_h \approx 52$ MeV — the Urban–Zhitnitsky value — as a convergence of *constraints*, not a measurement.

3. Σm_ν inheritance from Candidate C. The glueball horn keeps baryogenesis on the clock channel and therefore inherits the D-2 neutrino squeeze in full: normal ordering with $\Sigma m_\nu = 59$–64 meV, surviving in $\sim 1\%$ of the prior volume — one DESI tightening from death. *Status: live and near-term.*
4. Lattice $T_c/\Lambda_{\overline{\text{MS}}}$ at $N_f = 6$ (spec S-4). Pins Λ_h within $\sim 15\%$, deciding bare (28.5 MeV) vs UZ (52 MeV); disagreement with *both* falsifies the mirror-unified transmutation identification outright. The companion pure-glue discharge test $\rho_{\text{vac}}(L) - \rho_{\text{vac}}(\infty) \propto \chi_{\text{top}}/(m_G L)$ decides the load-bearing UZ conjecture itself. *Status: specified, external; computable with standard technology.*
5. Conditional LRD falsifier (P-LRD; trigger currently unmet). *If* LRD follow-up establishes that the seeds require SIDM gravothermal collapse (i.e. the astrophysical channels — supermassive stars, super-Eddington bursts, DCBH — are excluded), then in this framework: $\Lambda_h = 105$ MeV is dead as the DM scale and the surviving window is $\Lambda_h \approx 30$–52 MeV with baryon-boosted collapse. If LRDs are resolved astrophysically (the current mainstream expectation), this channel goes quiet and nothing here changes. The trigger condition is not currently met and may never be; the falsifier has not been appended to the pre-registration for that reason.

A sixth, related discriminator sits one level up: with transmutation replacing the visible-QCD amplitude hypothesis, the dissipation-partition parameter b is unpinned and becomes *measurable* (forecast $\sigma(\mu_0) \approx 0.05$ from DESI DR3 + Euclid separates $b = 0.2$ from $b = 1$ at $\sim 5\sigma$), giving a clean three-way origin discrimination: a measured $b \approx 0.206$ resurrects the visible-QCD hypothesis; any other value kills it in favor of transmutation or neither.

One robustness statement belongs alongside the falsifier list rather than in it. Because the hidden sector is cold ($\xi \approx 0.005$) and the glueball heavy ($m_G \approx 312$ MeV), cannibal freeze-out occurs at $T_{\text{vis}} \approx 3$ GeV, and both the free-streaming and cannibal-horizon cutoffs lie below $10^{-10}\,M_\odot$ — some eighteen orders of magnitude beneath the $\sim 10^8\,M_\odot$ subhaloes detected by gravitational imaging (e.g. JVAS B1938+666). The sector is CDM-like on all observable scales, as a *derived* consequence of its coldness rather than an assumption; substructure lensing does not constrain it, and the SIDM channel ($\sigma/m = 0.09$ cm^2/g) sits well below the ~ 1 cm^2/g scale at which subhalo evaporation becomes diagnostic in lensing — the binding self-interaction constraint remains the cluster bound of §6.

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8. Discussion and limitations

This section states plainly what is derived, what is matched, and what is conjectured.

claim	status
$1/\alpha_h(M_{\rm Pl}) = 83.2$ confines pure hidden $SU(3)$ at μ	derived
sensitivity $d\ln\Lambda/d\ln\alpha = 47.5$; scale naturalized, value log-selected	derived
mirror-unified $N_f = 6$ lands in the right decade with zero new continuous inputs	derived, and matched at the decade level
perturbative band 28–133 MeV, irreducibly $O(1)$-wide (near-conformal loop sensitivity)	derived
$\Lambda_h = 52$ vs 28.5 MeV discrimination	open — lattice $T_c/\Lambda_{\overline{\rm MS}}$ (spec S-4)
UZ mechanism $\rho \sim H\chi_{\rm top}/m_G$ (the law itself)	conjectural and contested — the one candidate compatible with nonlocal horizon degrees of freedom; lattice discharge test specified
$\alpha_{\rm eff} = 0.25$–0.6 for the glueball relic	derived, then matched to the required relic abundance
glueball abundance	selected, not derived: $Br \approx (0.4–3) \times 10^{-10}$ to $\sim 40\%$
$\sigma/m = 0.47/0.09/0.01$ cm²/g at $\Lambda_h = 30/52/105$ MeV	derived
LRD seeding at $\Lambda_h = 52$ MeV with $\times 30$ baryon boost	viability statement inside a bracket — not a prediction
three-way convergence on $\Lambda_h \approx 52$ MeV	convergence of constraints, one conditional — not a measurement

Limitations, in order of severity.

1. The law is conjectural. Everything that makes this a *dark-energy* proposal (as opposed to a natural 30–130 MeV hidden sector) rides on the contested UZ contact-term vacuum energy. We have argued it is uniquely aligned with the nonlocal horizon degrees of freedom that a volume-law count independently forces, and that it evades the Casini no-go, and we have specified a cosmology-free lattice test — but until that test (or an equivalent) is passed, the mechanism remains a conjecture and the route is incomplete.
2. The output band is irreducibly $O(1)$ perturbatively. The "zero new continuous parameters" statement is true of the mirror-unification *input*; the *output* carries a factor-~ 5 loop/scheme band because $N_f = 6$ sits near the conformal-window edge. The three-loop calculation was run precisely to tighten this and did not — it instead exhibited a spurious IR fixed point marking the failure of perturbative control.
3. The abundance dial. The glueball horn gives up ECCG-ADM's explained $\Omega_{\rm DM}/\Omega_b$ for a reheating branching ratio selected to $\sim 40\%$ within a 10^{-10} window. The mass–coupling sector is natural; the abundance is not explained.
4. Sector proliferation. The proposal adds a third hidden sector to the framework (the high-scale $SU(3)_H$ at 6×10^{12} GeV, the dark-force V/φ sector, and now mirror QCD at ~ 50 MeV) — a real cost against the economy claim, unless the mirror sector can be related to existing structure. Against this stands the strongest form of the economy accounting: $\alpha_h(M_{\rm Pl}) = \alpha_s(M_{\rm Pl})$ (no new coupling) + $N_f = 6$ (discrete) + $Br \sim 2 \times 10^{-10}$ (one small number) + the clock jointly deliver $\rho_{\rm DE}$'s scale, $\Omega_{\rm DM}$, $m_{\rm DM}$, η_B (given Σm_ν), and — via $\mu^3 = (6\pi\Omega_{X0}/u_0)\, a_0 M_P^2$ — the MOND acceleration scale a_0 [36], connecting the Planck-scale coupling to galaxy rotation curves in four steps. Against ΛCDM's two unexplained numbers (Λ, $\Omega_{\rm DM}$), that is a real economy, with the Br dial and the UZ conjecture as the residuals.
5. Mutual exclusivity with the visible-QCD origin. The transmutation origin and the alternative visible-QCD hypothesis ($b_{\rm QCD} = 0.2062$), developed in the companion cosmology manuscript, are mutually exclusive and distinguishable (§7, item 6); only one can be right, and current growth data ($\mu_0 = 0.04 \pm 0.22$) do not yet decide.
6. The LRD bracket is exponentially sensitive. The seeding result of §6.2 spans six orders of magnitude across its systematic bracket and enters the pincer only conditionally. We have deliberately phrased it as a viability

statement and kept the corresponding falsifier un-triggered.

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9. Conclusion

Dimensional transmutation converts SEDE's deepest fine-tuning — the origin of $\mu = 28.5$ MeV — into, at worst, a log-natural coupling specification and, in the mirror-unified version, a construction with no new continuous input at all. The score is: a zero-parameter decade-level success for the scale (band 28–133 MeV, irreducibly $O(1)$-wide perturbatively, with a specified lattice closer); one contested load-bearing conjecture for the law (UZ), aligned with the nonlocal horizon degrees of freedom that a volume-law count independently requires and equipped with a cosmology-free lattice discharge test; a glueball dark-matter branch whose mass and coupling are natural but whose abundance is a 10^{-10}-level selection; and a set of near-term, mutually independent probes — cluster SIDM, a conditional LRD bracket, the lattice ratio, DESI Σm_ν, and the growth-measured b — that either converge on $\Lambda_h \approx 52$ MeV or take the identification apart. The proposal is presented at exactly that strength and no more.

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10. Reproducibility

All numerical claims in this paper are reproduced by the following scripts in this repository (`https://github.com/spsingularity/mu-transmutation-glueball`, `src/`; a tagged release is archived at Zenodo, DOI 10.5281/zenodo.21525532):

- `src/mu_transmutation_check.py` — §2 coupling check and sensitivity; §3.1 one-loop N_f scan; §4 UZ coefficient target.
- `src/mu_transmutation_steps.py` — §3.2 two-loop + threshold band; the b-freed sum rule and measurability forecast; §5.4 subcomponent ξ boundaries and the m_X-shift window.
- `src/mirror_qcd_3loop.py` — §3.3 three- and four-loop running, the spurious IR fixed point, and the 28–133 MeV band.
- `src/glueball_simp_relic.py` — §5 decoupled-cannibal freeze-out, α_{eff} and ξ/Br bands, ΔN_{eff}, and the §6.1 σ/m ladder.

- `src/glueball_lrd_seeding.py` — §6.2 gravothermal seeding calculation (halo mass function, concentration tail, collapse times, seed abundances).

Source assessments (dimensional transmutation, glueball-DM, and LRD seeding) are documented in the private research hub.

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This research received no external funding. The author, an independent researcher, declares no competing interests. Ethics approval is not applicable: this work is theoretical and involved no human participants, no human data or tissue, and no animal subjects.

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11. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude Opus 4.x (Anthropic) in order to draft and edit the manuscript, to develop and cross-check the analysis code, and to assist with literature searches. After using this tool the author reviewed and edited the content as needed — including checking every literature reference against its source — and takes full responsibility for the content of the published article. No AI tool is an author.

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